



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

DEPARTMENT OF STATISTICAL SCIENCES
Second Cycle Degree in Statistical Sciences

Visualizing ML classification results: recent graphical tools and novel extensions

Presented by:

Lorenzo Lazzari

Supervisor:

Prof. Laura Anderlucchi

14/07/2023

Dissertation Discussion

1

Overview

Recently proposed set of Graphical Tools for classifiers
[by J. Raymaekers and P. J. Rousseeuw (2022)]

analysis &
comprehensive review

extension of the
existing framework

Visualize ***classification results on a labeled set of data***

Train Set

or

Test Set

→ Understand classifier performance (in specific task)

→ Understand structure of the data

Model optimization
Model comparisons
Conscious deployment
.....

2

Silhouette Plot

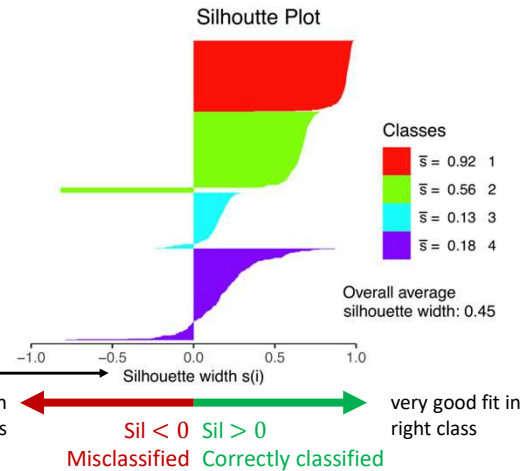
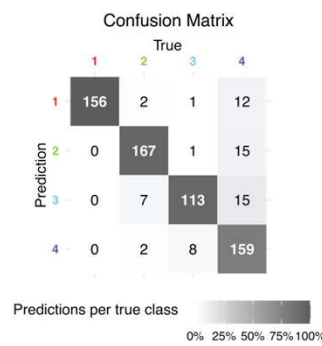
Interpreting posterior probabilities

$\hat{P}_i(g)$ *estimated posterior probability*

$g=1,2,3,4$ classes, with g_i true class for observation i

$$PAC(i) = \frac{\tilde{P}_i}{\hat{P}_i(g_i) + \tilde{P}_i}$$

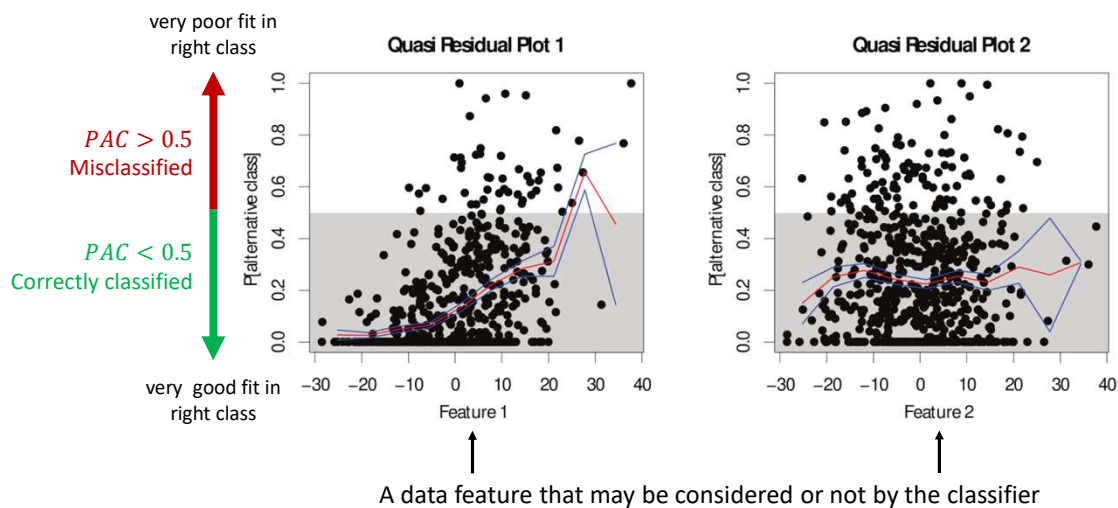
$$S(i) = 1 - 2PAC(i)$$



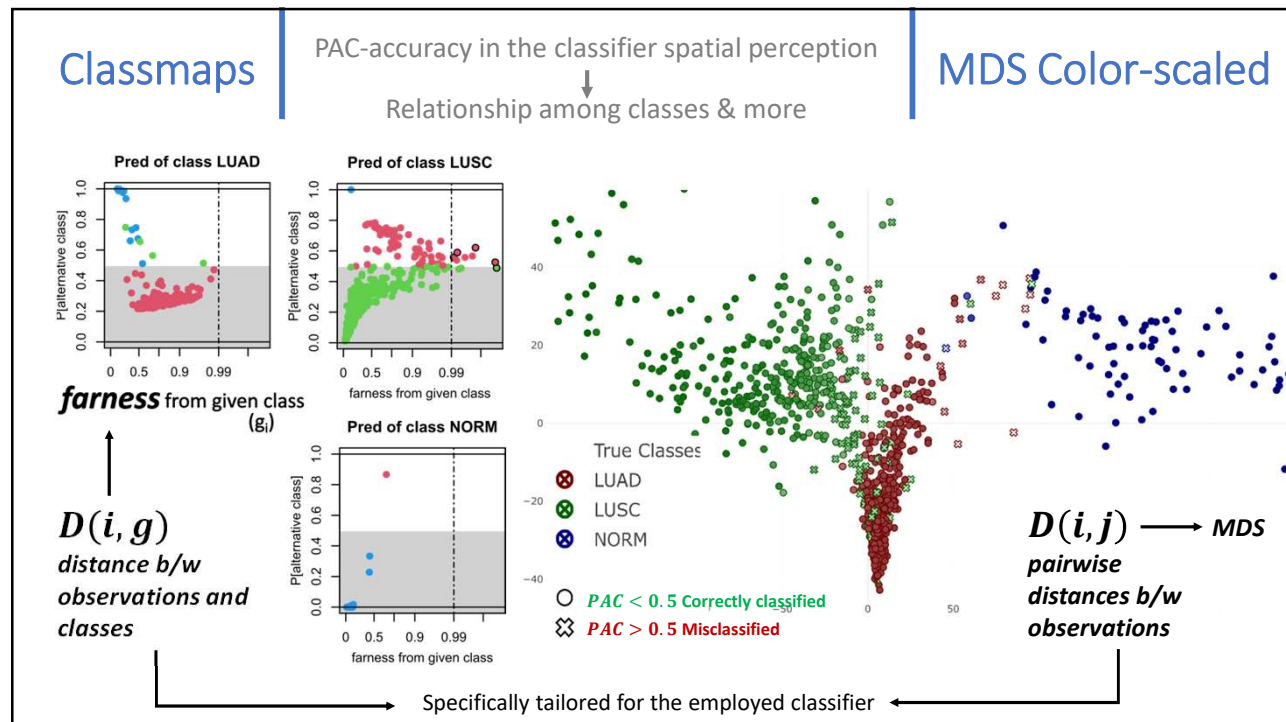
3

Quasi-residual plots

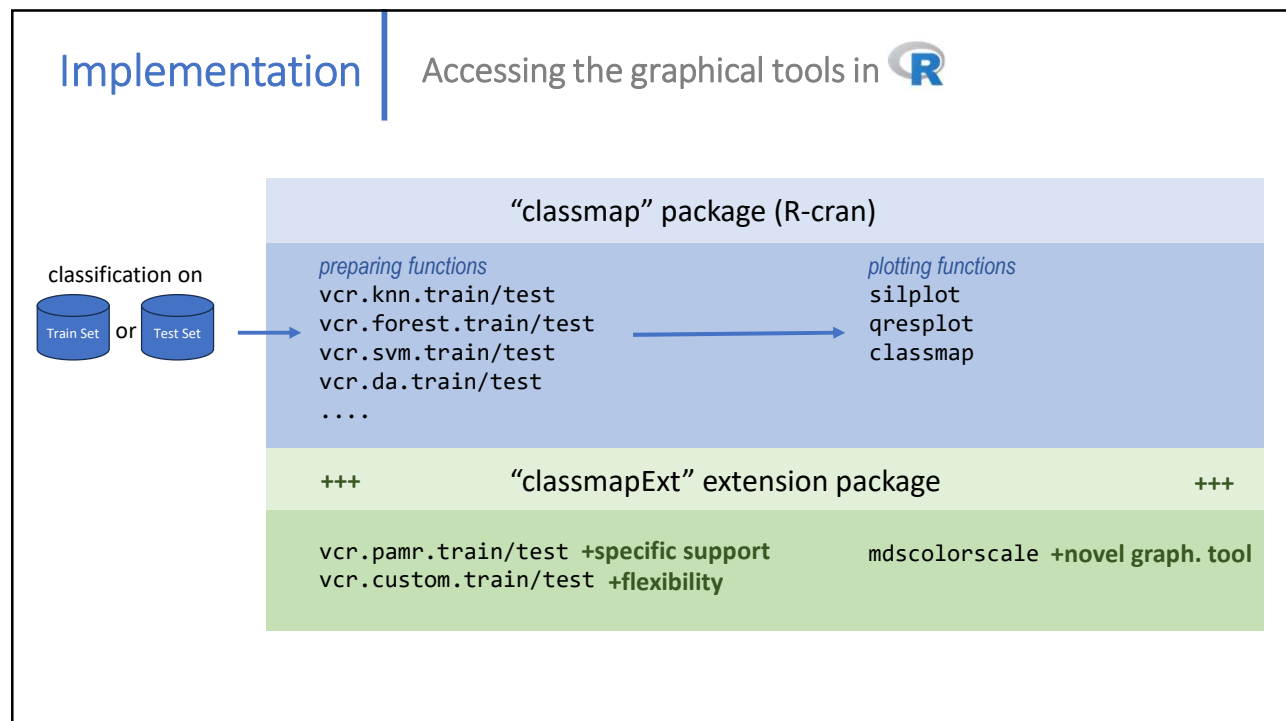
Exploring associations b/w covariates and PAC



4



5



6

Application Case

Visualizing gene expression-based cancer classification

